CYBER SECURITY ATTACK ANALYSIS

40K

Total Attacks Recorded

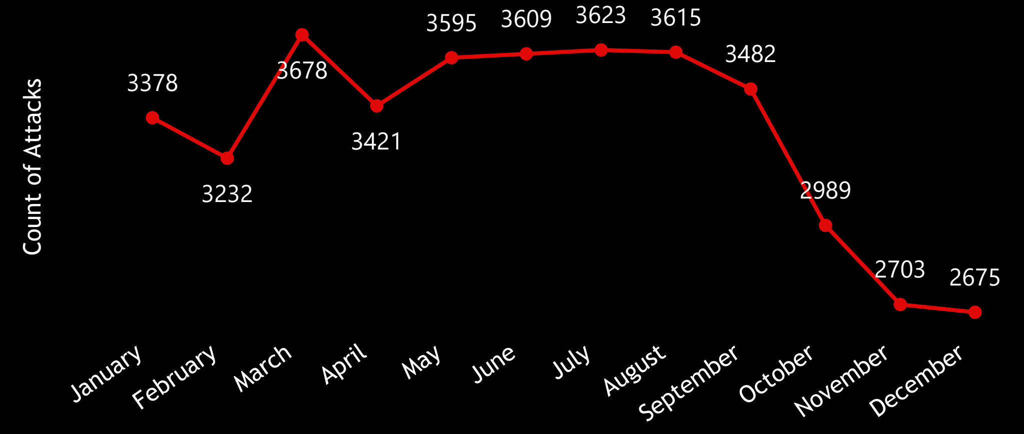
2020

2021

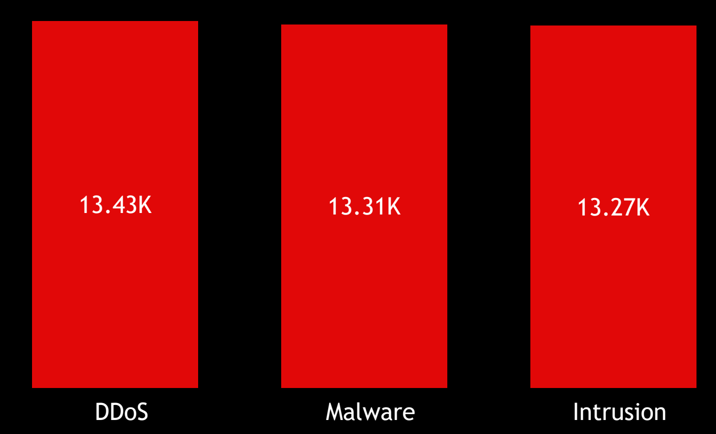
2022

2023

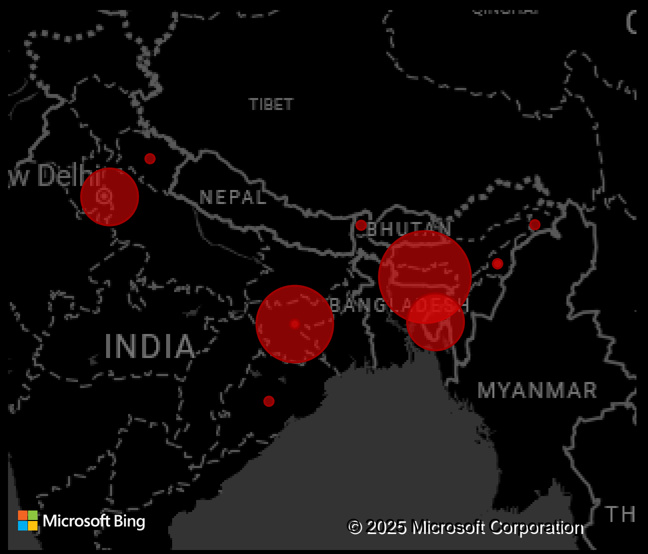
Attack trend over time by month



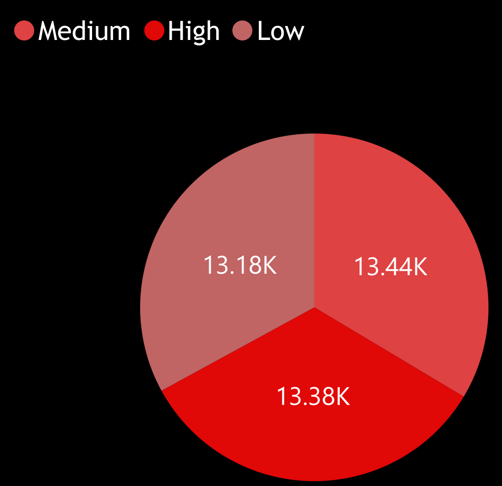
Most frequent attack type



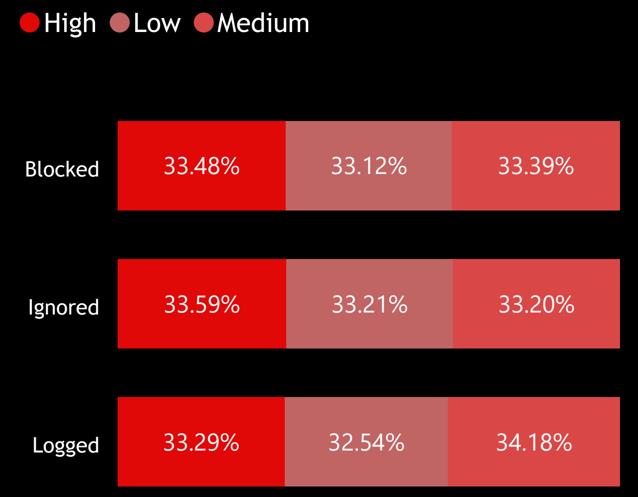
Region with highest attacks



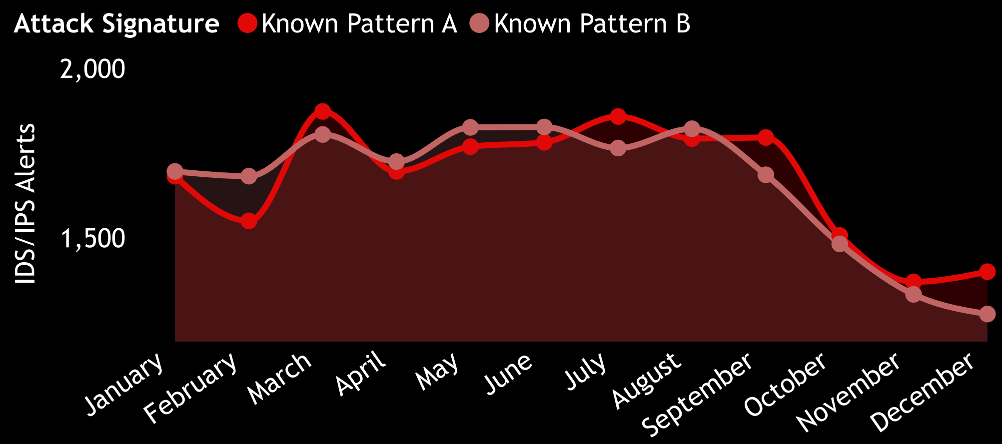
Severity of Attacks



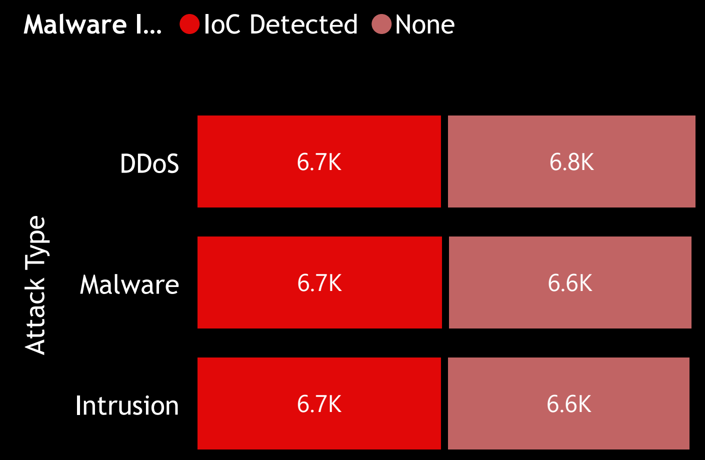
Action taken by Severity of attack



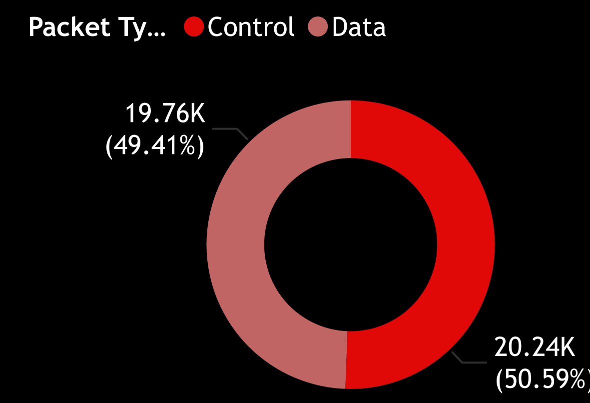
Trend of Security Alerts Over Time



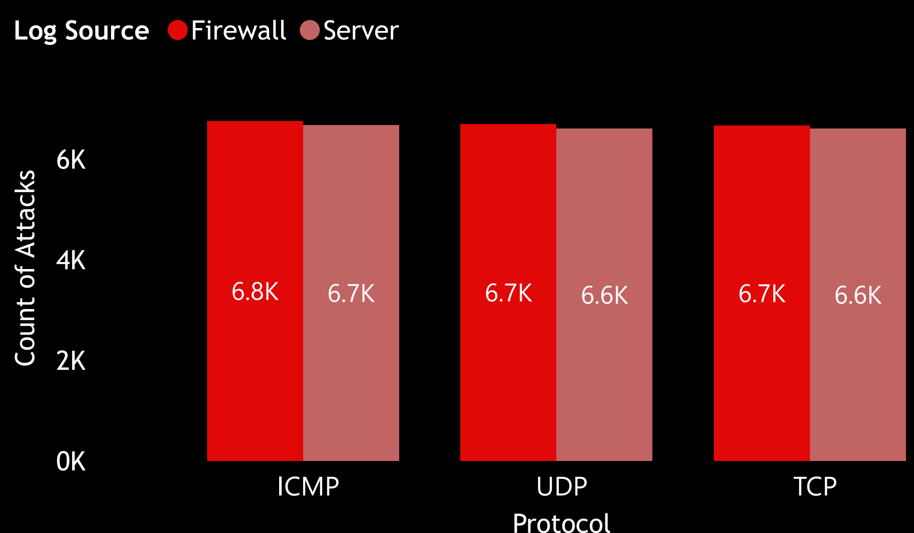
Proportion of Data vs. Control Traffic



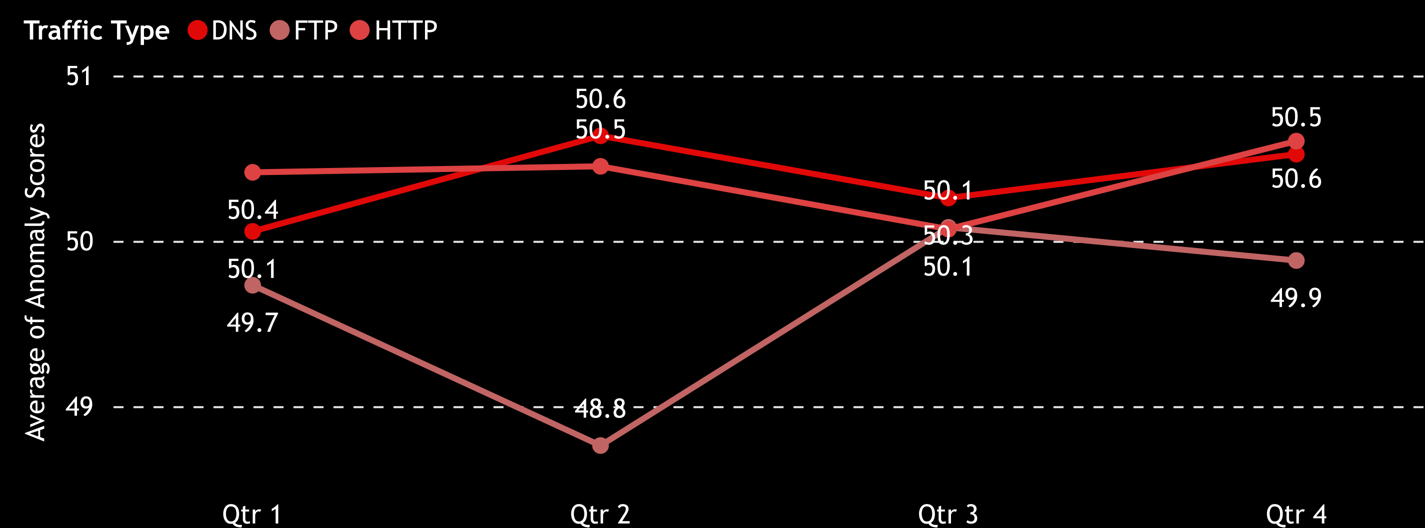
Attack Methods by Protocol and Log Source



Attack Types and Their Malware Associations



Unusual Network Activity Over Time



Cyber Security Attack Analysis Report

Problem Statement:
Organizations need to monitor cyber-attacks to identify trends, common attack types, and the effectiveness of security responses. This dashboard provides an analysis of attack types, severity levels, and regional distribution.

Technologies Used:

- **SQL**: Data extraction and transformation
- **Power BI**: Data visualization and analysis

Key Insights:

- **Total Attacks**: **40K cyber-attacks** recorded, with peaks in mid-year months.
- **Most Common Attack Types**: DDoS, Malware, and Intrusion are the top threats, each exceeding **13K occurrences**.
- **Severity Levels**: High, medium, and low severity attacks are almost evenly distributed.
- **Actions Taken**: The majority of high-severity attacks were blocked, while medium-severity ones had a significant number ignored or logged.

Recommendations:

1. **Strengthen Defense Against Top Attack Types** – DDoS, malware, and intrusion attacks are the most frequent. Invest in advanced firewalls, AI-based threat detection, and endpoint protection solutions.

1. **Regional-Specific Security Measures** – Focus cybersecurity resources on regions with the highest attack counts by implementing geo-blocking and local threat intelligence solutions.

1. **Improve Incident Response Time** – Since attacks vary in severity, prioritize quick responses to **high** and **medium-severity** attacks while automating detection for low-risk incidents.